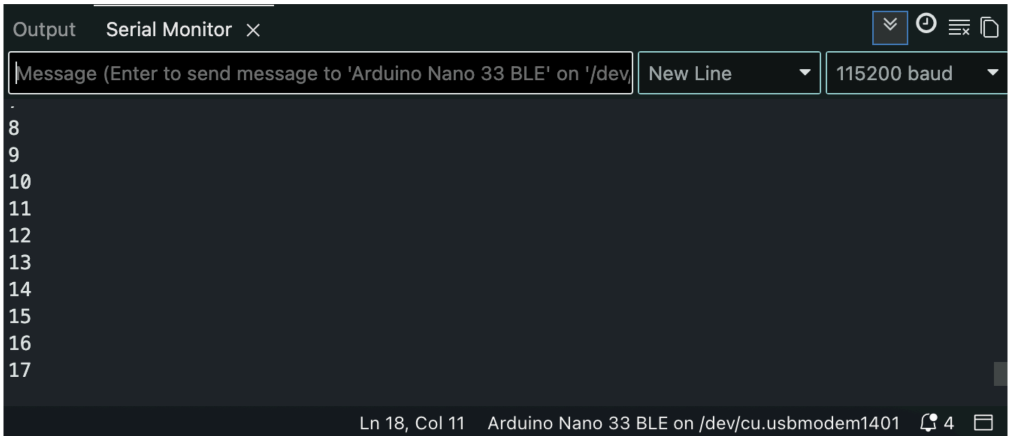
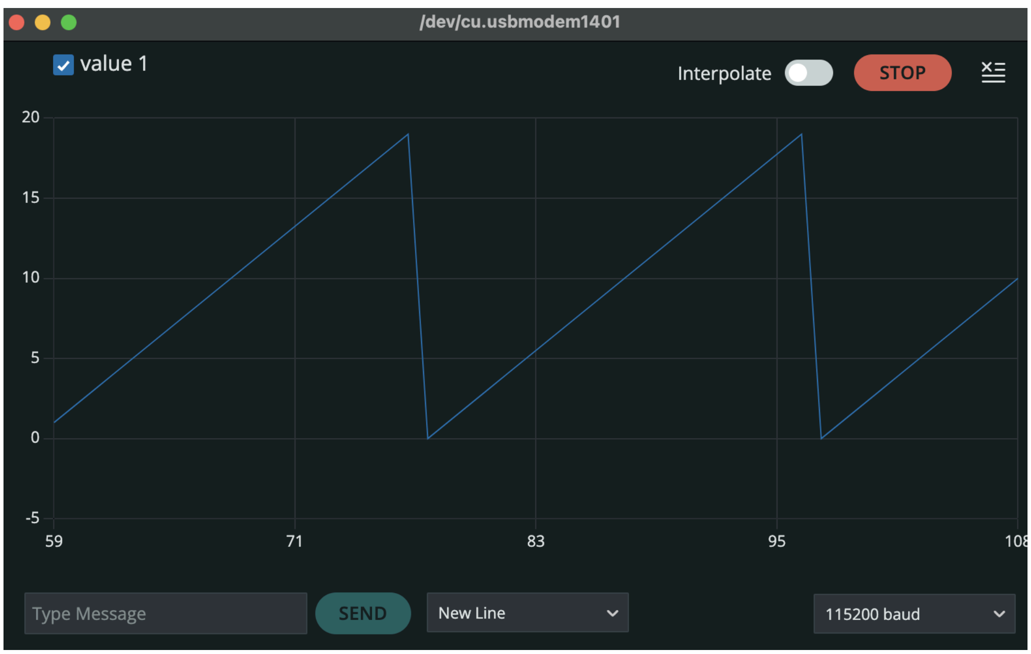
Task 4:

Serial Monitor:

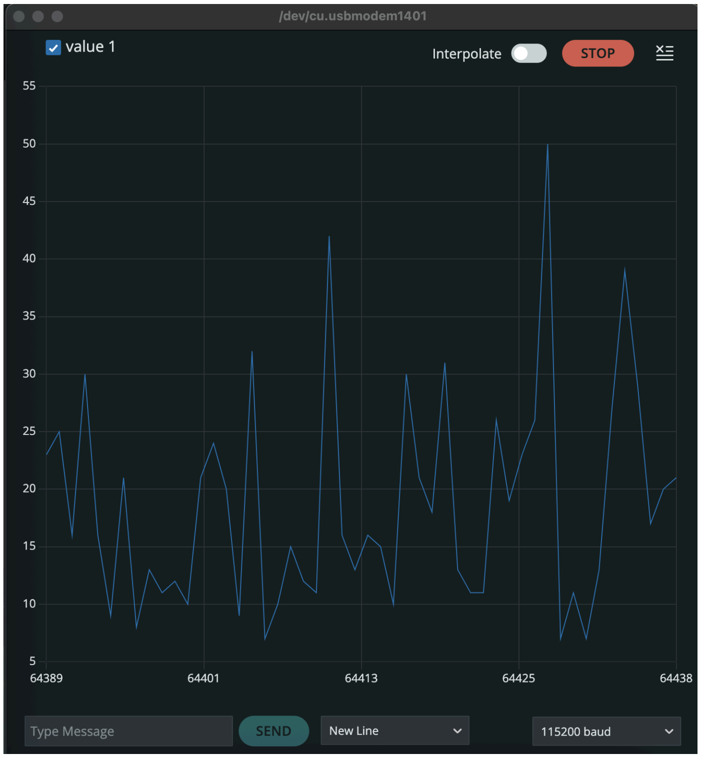


Serial Plotter:

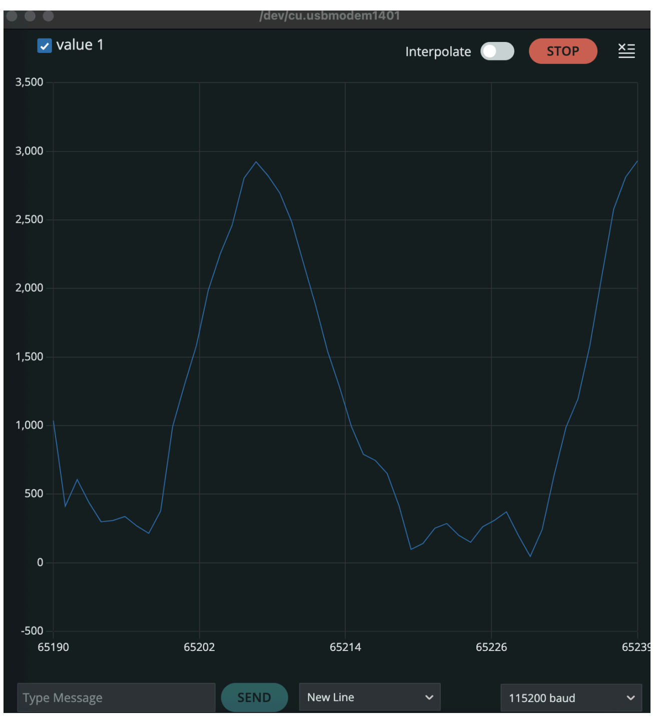


Task 5:

Serial Plotter Quiet:

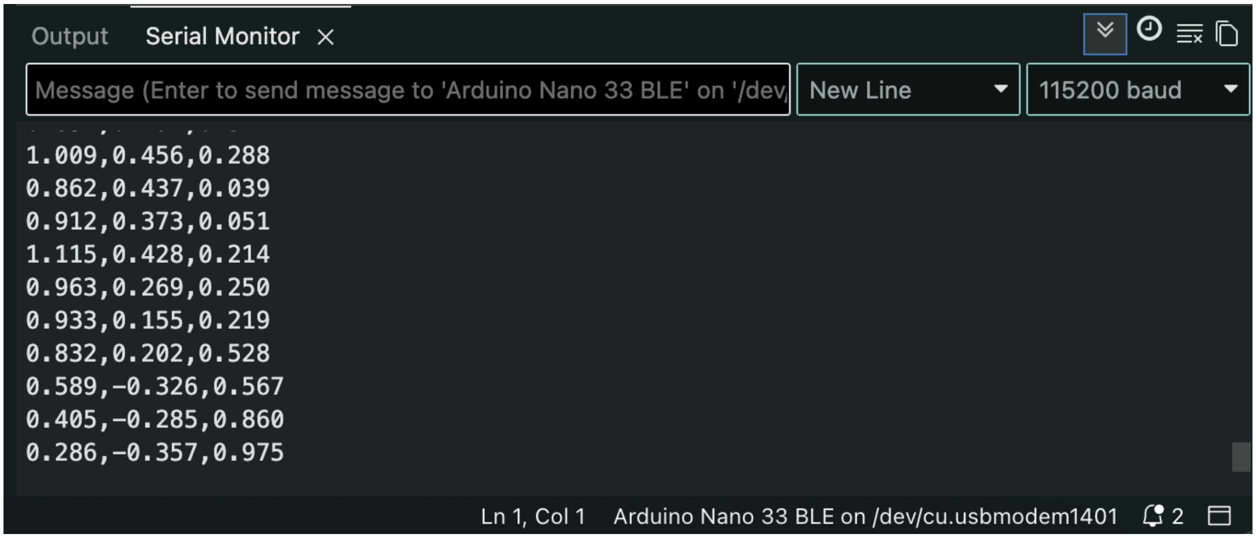


Serial Plotter Loud:

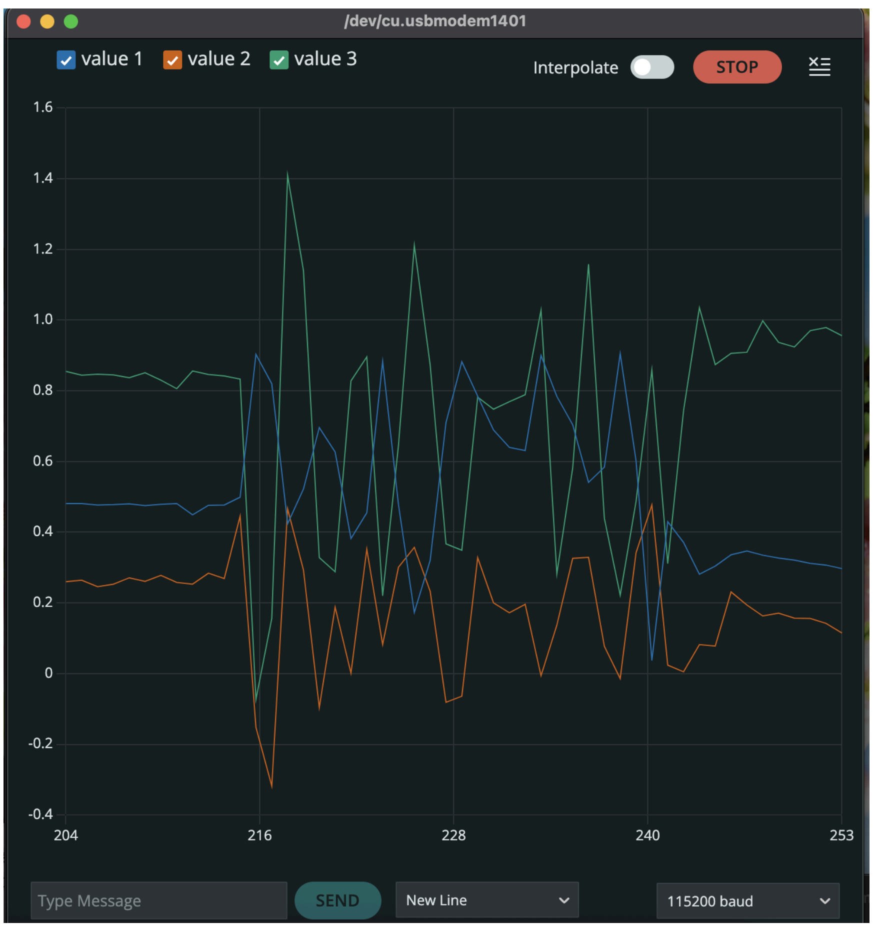


The loud plot has significantly higher amplitudes (~ 3,000) compared to the quiet plot which has amplitudes of ~40-50.

Task 6A Serial Monitor:



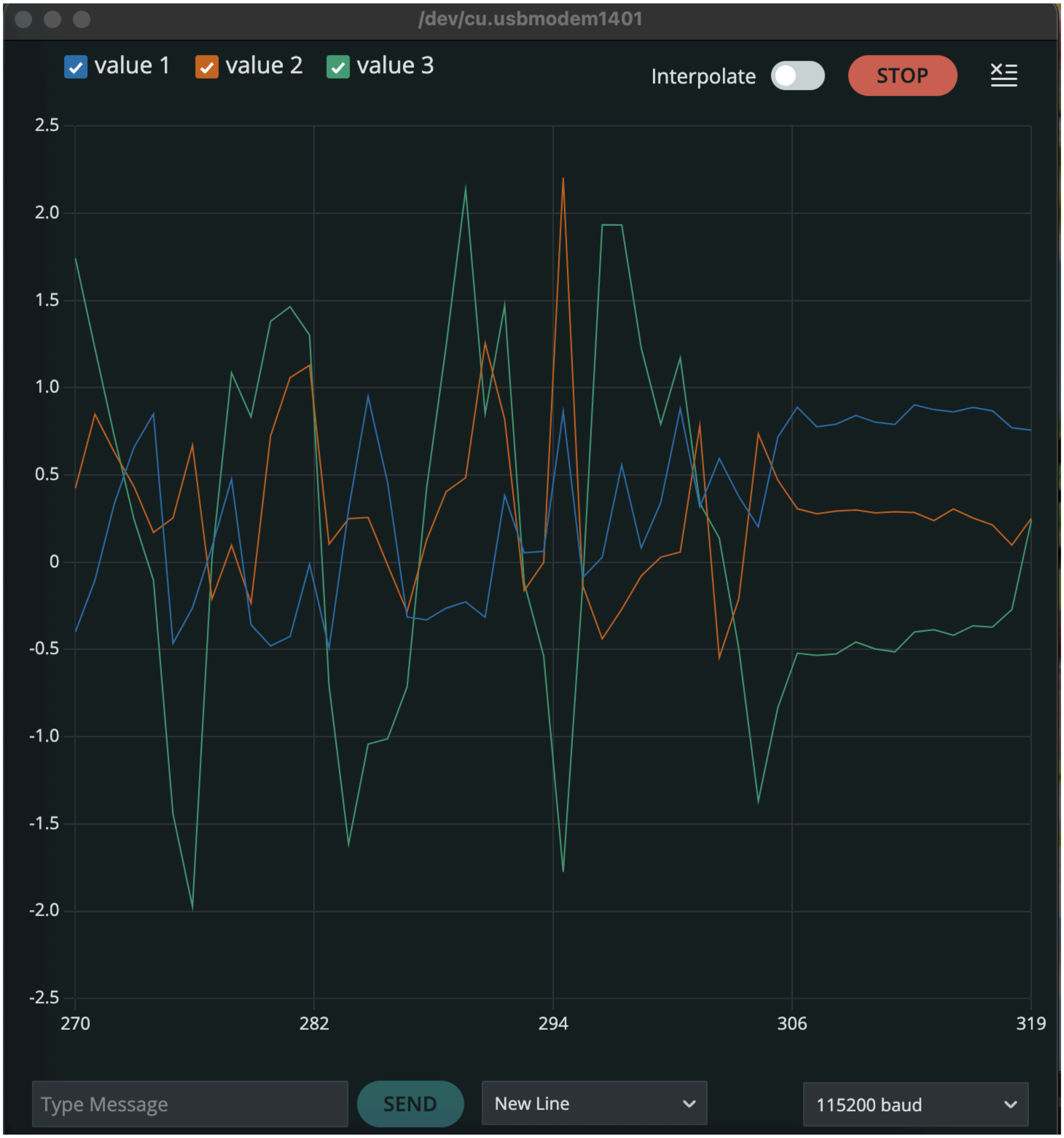
Task 6A Serial Plotter:



The movement that produced the most change was drawing a figure “8” with the arduino. I think was because there was inclusion of up, down, left, and right movements.

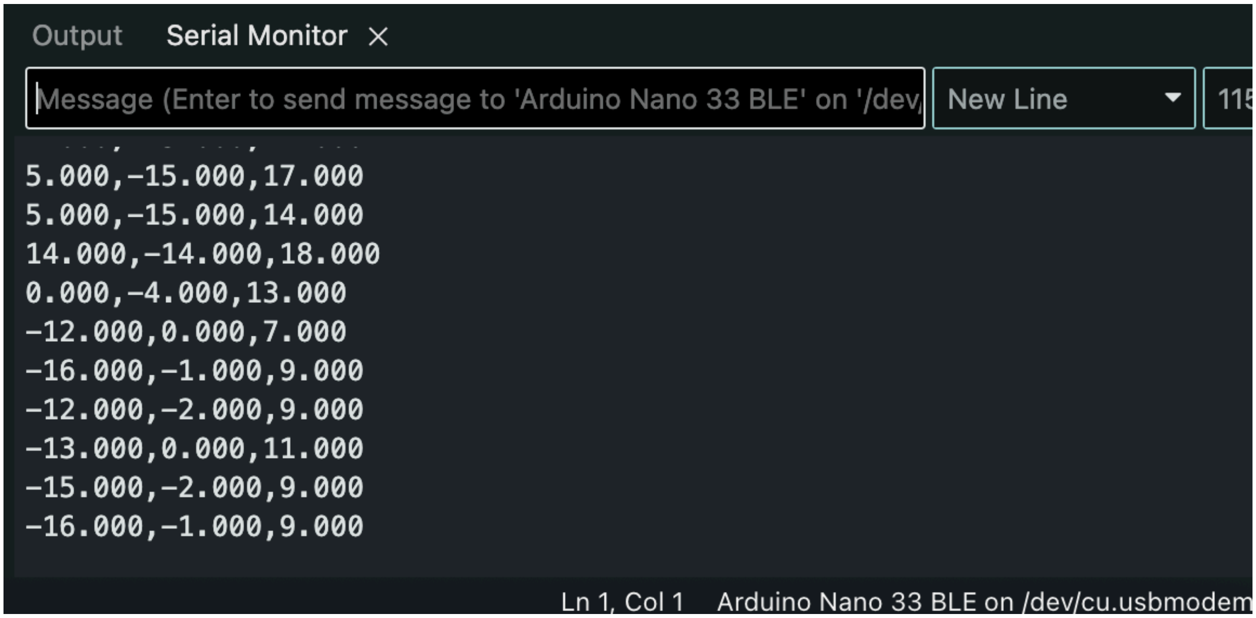
Task 6B Serial Monitor:

Task 6B Serial Plotter:



The movement that produced the most change was vigorously shaking the board front, back, and then up, and down.

Task 6C Serial Monitor:



Task 6C Serial Monitor:

Putting the arduino nano on top of my laptop produced the greatest change with the serial monitor values and the serial plot signal graphs.

Task 7 Normal Temperature Serial Monitor:

Output Serial Monitor

Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/ | New Line | 115200 baud

```
Temperature (C): 25.10 | Humidity (%): 46.55
Temperature (C): 25.09 | Humidity (%): 46.49
Temperature (C): 25.07 | Humidity (%): 45.30
Temperature (C): 25.05 | Humidity (%): 44.86
Temperature (C): 25.03 | Humidity (%): 44.57
Temperature (C): 25.05 | Humidity (%): 44.05
Temperature (C): 25.05 | Humidity (%): 44.00
Temperature (C): 25.00 | Humidity (%): 43.45
Temperature (C): 25.02 | Humidity (%): 43.40
Temperature (C): 25.03 | Humidity (%): 43.72
```

Task 7 Increased temperature and humidity serial monitor:

```
Temperature (C): 27.64 | Humidity (%): 86.86
Temperature (C): 27.42 | Humidity (%): 86.95
Temperature (C): 28.39 | Humidity (%): 89.44
Temperature (C): 28.75 | Humidity (%): 90.97
Temperature (C): 28.55 | Humidity (%): 91.26
Temperature (C): 28.02 | Humidity (%): 90.82
Temperature (C): 27.46 | Humidity (%): 89.96
Temperature (C): 28.13 | Humidity (%): 90.82
Temperature (C): 28.35 | Humidity (%): 91.31
Temperature (C): 28.27 | Humidity (%): 91.27
```

Lightly breathing on the arduino boosted the temperature and humidity values.

Task 8:

```
Pressure (kPa): 100.696 | Temperature (C): 22.57
Pressure (kPa): 100.692 | Temperature (C): 22.57
Pressure (kPa): 100.695 | Temperature (C): 22.57
Pressure (kPa): 100.694 | Temperature (C): 22.57
Pressure (kPa): 100.693 | Temperature (C): 22.57
Pressure (kPa): 100.695 | Temperature (C): 22.57
Pressure (kPa): 100.693 | Temperature (C): 22.57
Pressure (kPa): 100.696 | Temperature (C): 22.57
Pressure (kPa): 100.695 | Temperature (C): 22.56
Pressure (kPa): 100.697 | Temperature (C): 22.56
Pressure (kPa): 100.697 | Temperature (C): 22.55
```

Task 9A:

```
Proximity: 240
Proximity: 241
Proximity: 240
Proximity: 238
Proximity: 0
Proximity: 0
Proximity: 0
Proximity: 0
```

Task 9B:

```
UP
DOWN
LEFT
RIGHT
LEFT
RIGHT
UP
DOWN
LEFT
```

Task 9C:

Baseline room lighting:

```
63,65,74,188
63,65,74,187
62,71,71,178
58,69,69,168
58,70,70,169
```

Colored Room lighting:

- Red:

```
295,309,326,863
297,312,329,872
294,309,326,862
296,311,327,864
298,313,330,872
302,317,334,883
298,313,330,872
```

- Blue:

```
5,30,79,109
5,30,79,110
5,29,76,106
5,29,78,106
5,29,78,108
5,30,78,108
```

The change that was the most drastic was when I put my phone displaying a specific color near the sensor.

Task 10:

```
raw,mic=12,clear=38,motion=1.4437,prox=56
flags,sound=0,dark=1,moving=1,near=1
state,QUIET_BRIGHT_STEADY_FAR
```

```
raw,mic=442,clear=176,motion=1.6461,prox=242
flags,sound=1,dark=0,moving=1,near=1
state,NOISY_BRIGHT_MOVING_NEAR
```

```
raw,mic=12,clear=20,motion=0.1610,prox=32
flags,sound=0,dark=1,moving=0,near=1
state,QUIET_DARK_STEADY_NEAR
```

```
raw,mic=27,clear=116,motion=0.2015,prox=241
flags,sound=1,dark=0,moving=0,near=0
state,NOISY_BRIGHT_STEADY_FAR
```

I selected my thresholds by first setting random values and watching the serial monitor fluctuate and then fine tuning the values. For audio, I first tested the quiet room microphone settings and then clapped to observe how the value would change. For light, I put the board in my room, and then turned the lights off. For proximity, I put nothing near the sensor, then would put my hand or object close to the board, and then logged the values.

Task 11:

```
raw,rh=39.56,temp=23.98,mag=3.74,r=49,g=53,b=64,clear=163
flags,humid_jump=0,temp_rise=0,mag_shift=0,light_or_color_change=0
event,BASELINE_NORMAL
```

```
raw,rh=90.89,temp=26.58,mag=58.99,r=49,g=53,b=64,clear=163
flags,humid_jump=1,temp_rise=1,mag_shift=1,light_or_color_change=0
event,MAGNETIC_DISTURBANCE_EVENT
```

```
raw,rh=85.85,temp=25.34,mag=35.69,r=49,g=53,b=64,clear=163
flags,humid_jump=1,temp_rise=0,mag_shift=0,light_or_color_change=0
event,BREATH_OR_WARM_AIR_EVENT
```

I selected the thresholds first by using the serial monitor to watch the normal baseline values. Then I chose thresholds that are slightly above normal so I would see the changes. The cooldown logic prevents the same event from printing too fast.